Final Exam to 2025/2026

Linear Algebra (I) :

Secod year of (Mathematic+Physics)

Question 1:

a- Find the det (A) of the Matricx $A = \begin{bmatrix} -7 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 5 \end{bmatrix}$

b-Find the solution for linear system by Gaussian Elimination method

$x - 2y + Z = 7$

$3x - 2y + 2Z = 14$

$2x - y + 4Z = 17$

c- Find the Matrix $(3A - B)$ *where* $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$ *and* $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$?

d-If A= $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ *prove that* $A.A^{-1} = I$?

Quesetion 2:

a – Find an inverse B^{-1} If *the Matrix* $B = \begin{bmatrix} 6 & 0 & 0 \\ 0 & -10 & 0 \\ 0 & 0 & 6 \end{bmatrix}$?

b-If $u = \{(1,2,0), v(2,1,0)\}$ *two vectors in* R^3 *find* $u + v$ *and* $u \times v$?

c- Find the Angle θ *between the vectors* $u = (1,-1,1)$*and* $v =$ $(1,1,-1)$?

d-Find the valueces of x, y ,z to make Symmetric Matrix $A =$ $\begin{pmatrix} 4 & 10 & 3x \\ 8-2y & 2 & 2z \\ 1 & 6 & 2x+1 \end{pmatrix}$?

Midterm Exam
Linear Algebra

Q1 If $A = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$

Find the Matrix $A+B$.

Q2 If $A = \begin{bmatrix} 1 & x & 2 \\ 5 & 6 & -1 \end{bmatrix}^T = \begin{bmatrix} 1 & 1- \\ -3 & 6 \\ 2 & y \end{bmatrix}$

Find x and y

Q3 Find det of A If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 5 \\ 0 & 0 & 4 \end{bmatrix}$

Q4 Find Inverse of the Matrix A^{-1}

If $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$?

Q5 If $\begin{vmatrix} y & 5 & 13 \\ 7 & 3y-1 & 0 \\ y+1 & 0 & 0 \end{vmatrix} = 0$ find the value of y

Q6 Find an Equivalent Matrix B and the Rank $r_{21}(-1)$, $r_{31}(-1)$.

If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$